



LABORATORY 7. *Drosophila* CLASS DATA REVIEW: GENE MAPPING AND HYPOTHESES TESTING

LEARNING OUTCOMES

At the end of this lab practice, students should be able to:

- Determine the genetics model of inheritance of four genes based on real cross data.
- Propose a scientific hypothesis (dihybrid cross) and statistically test its validity.
- Map three genes on a chromosome based on real cross data.

INTRODUCTION

The main aim of this lab exercise is to analyze and discuss the outcome of a particular cross and its further generations (F_1 and F_2). Additionally, students will get familiar and comfortable at solving genetics problems and analyzing the different epistatic interactions between genes.

Based on the previous labs (#1, 2, 3 and 4), students should be familiar on the methods and

analysis to determine the genetics mode of inheritance of different mutations by using the phenotypic classes and their ratios on the F_1 and F_2 generations. Additionally, students must be familiar with the potential epistatic interactions between some genes related with eye color variation (lab # 5). Let's use this previous knowledge and new skills to answer questions regarding your real-case genetics experiment performed in this lab.

You have been analyzing a *Drosophila* cross between P_1 yellow body, white eyes, and miniature wing females with a P_2 sepia eyes males (both homozygous lines). During a period of four weeks, you collected and recorded F_1 and F_2 data that will allow you to test some hypotheses about the inheritance and to map three genes on a chromosome.

Please **read each paragraph and question carefully**. Your understanding of the concepts is key for your success in your assignments.



P_1 : Females (yellow body, white eyes, miniature wing mutant)

X



P_2 : Males (sepia eyes)



GENOME WIDE ASSOCIATION STUDIES

100

Three-point mapping and genome-wide association studies (GWAS) both aim to identify genomic regions associated with a trait. In three-point mapping, researchers analyze recombination events in controlled crosses to determine the location of a gene relative to known markers. In contrast, **GWAS examines natural populations** and tests whether specific SNPs (Single Nucleotide Polymorphism) occur more frequently in individuals with a particular phenotype. Although the approaches differ, both rely on the same principle: genetic markers located close to a causal gene tend to be inherited together because recombination between them is infrequent. As a result, both methods can identify genomic regions linked to a trait, helping researchers narrow down the location of genes influencing phenotypes.

PROCEDURE

The tables indicating the outcome (i.e., offspring number) of the cross performed **in your lab section** are available in front of the lab.

Please analyze those results carefully, as they are the basis to complete the lab assignment. To facilitate your understanding of the concepts and solving your assignment, please consider the following key aspects:

- The cross involves **four** genes. **Two** different genes are responsible for **white** and **sepia** eyes.

- Identify the location of genes on the chromosomes. That is, take into consideration sex-linked vs. autosomal genes. The F_1 results will provide you hints in this regard.

- Due to an epistatic interaction, the F_2 results of this cross yields 12 phenotypic classes (instead 16 classes).

- For the three-point chromosome mapping of yellow, white, and miniature, you should add up some of the F_2 phenotypic classes together. That is, you should end up with a total of eight (8) phenotypic classes (instead of 12). Parental offspring classes ($n=2$), single recombinant classes ($n=4$) and double recombinant classes ($n=2$).

For the gene mapping and χ^2 test, you will use the **"Three Point Mapping"** virtual tool available on Canvas or here (www.ampossot.com/3point). Follow the indications on the app, complete the tables and perform your analysis.

During the second part of the lab, you will analyze data from a simulated GWAS experiment in a human population. To do so, you will use the **"GWAS virtual tool"** available at www.ampossot.com/gwas. Follow the indications in the lab manual to complete the analysis.

**IMPORTANT: FILL THE TABLES WITH THE DATA PROVIDED BY YOUR TAs**

Your lab section (e.g., Tuesday morning): _____

Parental flies (homozygous, pure lines)**P₁:** Females, yellow body, white eyes, miniature wings.**P₂:** Males, sepia eyes**F₁ results**

	Males	Females	Total
	Yellow, white, miniature	Wild-type	
Total			



F₂ results

Phenotype	Sex	Amount	Total
WT	F		
	M		
y, w, m	F		
	M		
y	F		
	M		
w, m	F		
	M		
y, w	F		
	M		
m	F		
	M		
w	F		
	M		
y, m	F		
	M		
se	F		
	M		
y, se	F		
	M		
m, se	F		
	M		
y, m, se	F		
	M		



ASSIGNMENT # 7

DEADLINE: RETURN AT THE END OF THE LABORATORY SESSION**OPTION 1:** Deposit a physical copy in the cart (front of the lab)**OPTION 2:** Submit an electronic copy (i.e., PDF file) through the LMS platform (Canvas)

NAME: _____ STUDENT #: _____ DATE: _____

GENE MAPPING BASED ON YOUR COLLECTED DATA

1. Using the class data, construct a linkage map of the yellow (**y**), white (**w**) and miniature (**m**) genes. **Show your work.** You will use the **“Three Point Mapping” virtual tool** available on Canvas or here www.ampossot.com/3point. Follow the instructions in the app and use the table below to summarize the observed data.

Hint: Are you wondering about the classes with the **sepia** phenotype? You have to omit that trait and consider only the **y**, **w** and **m** genes. For example, the number of flies in the class “**sepia**” should be added to the class “**WT**” (because they have the wild-type alleles (**y⁺ w⁺ m⁺**) for these genes). Follow the same rule to the remaining **three** classes showing the sepia trait.

Then, identify parental (**PR**) (larger classes) and double recombinant (**DR**) (smaller classes). The gene whose allele differs between the PR and DR classes is the middle gene.

Phenotype	Observed number	Genotype (F1 female gametes)	*Type of gamete (PR, SR or DR)
Wild type			
Yellow, white, miniature			
Yellow			
Miniature, white			
Yellow, white			
Mini			
White			
Yellow, mini			
Total			

2. Gene order: _____



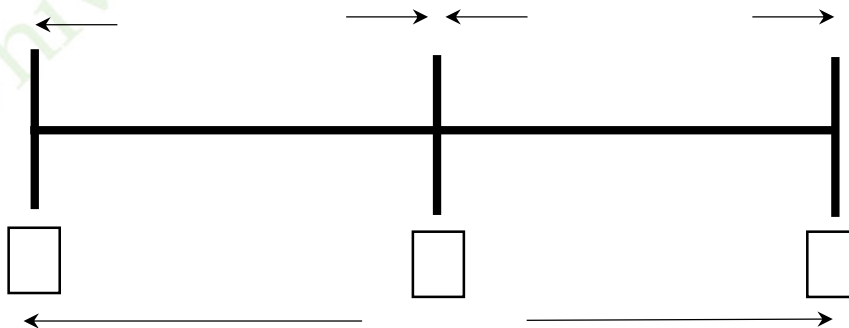
Calculate the genetic distances (show your work and units):

3. $y-w=$

4. $w-m=$

5. $y-m=$

Summary distances:





To calculate Coincidence and Interference, please review the formulas:

$$C = \frac{\text{Obs. Freq. DR}}{\text{Exp. Freq. DR}}$$

$$I = 1 - C$$

6. C: _____

7. I: _____

8. Please provide a brief explanation of the Coincidence and Interference values that you calculated. Is there any "interference"? What does it mean?

**X² BASED ON YOUR COLLECTED DATA**

9. Set up a χ^2 to test the hypothesis that the yellow (**y**) (body color) and miniature (**m**) (wing size) are **recessive mutations** that assort **independently**. The **“Three Point Mapping” virtual tool** available on Canvas (www.ampossot.com/3point) allows you to perform such test after the mapping task.

Hint: Are you wondering about the classes with the **other** phenotypes? You must omit those traits and consider only the **y** and **m** genes. For example, the number of flies in the class “**sepia**” should be added to the class “**wild**” (because they are **y⁺ m⁺** for these genes). Follow the same rule to the remaining classes.

Expected ratios under H ₀	Phenotype	Observed number	Expected number
	Wild type		
	Miniature wing		
	Yellow body		
	Yellow body, miniature wing		
	Total		

χ^2 Calculated: _____ Degrees of freedom: _____

p=0.05): _____

Reject or fail to reject H₀: _____

10. After the χ^2 test, what is your conclusion regarding the mode of inheritance of the **y - m** genes? Please provide a brief explanation.



GENOME WIDE ASSOCIATION STUDY

Conceptual connection: Like three-point mapping, GWAS attempts to identify genomic regions associated with a phenotype. However, GWAS uses allele frequencies in populations rather than recombination frequencies from genetic crosses.

For the following exercise, you will use the data collected from the “GWAS virtual tool” (www.ampossot.com/gwas).

A cohort of human individuals was sequenced using mock next-generation sequencing data. Sequencing reads have already been processed into SNP genotypes. Your task is to identify which genomic region is associated with a hypothetical disease phenotype.

11. In this simulation, you are studying a group of individuals **affected** or **unaffected** by a disease. The goal is to identify genomic regions in which SNP alleles are significantly associated with disease status. If a SNP **increases** disease risk, the risk allele should be more common in affected individuals. Then, the null hypothesis should be that the risk allele frequency is the same in affected and unaffected individuals.

Answer the following questions in the app and record your answers

Step 1: Study Design Prediction

If a SNP is associated with increased disease risk, what pattern should you expect?

Select one answer

What is the null hypothesis for this SNP association analysis?

Select one answer

Data Summary

Individuals	Affected	Unaffected	Chromosomes	SNPs

12. Calculate risk allele frequencies for one candidate SNP.

SNP ID	Chromosome	Affected risk alleles	Affected total alleles	Unaffected risk alleles	Unaffected total alleles

Calculate the risk allele frequency in **affected** individuals (affected risk alleles / affected total alleles). Show your work



Calculate the risk allele frequency in **unaffected** individuals (unaffected risk alleles / unaffected total alleles). Show your work

13. The app will calculate the Odds ratio (OR). This is a statistical measure to estimate the strength of the association between a SNP and the disease. The larger the OR, the stronger the association between the SNP and the disease.

OR = 1 → no association

OR > 1 → allele enriched in affected individuals

OR < 1 → allele may be protective

Answer the following questions in the app and record your answers

What does an odds ratio greater than 1 suggest?

Select one answer

Fill the table below:

Group	Observed <u>risk</u> alleles	Observed non-risk alleles	Total alleles
Affected			
Unaffected			
Total			

Under the null hypothesis, risk alleles should be distributed randomly between affected and unaffected individuals. The **expected** values represent what we would observe if there were no association between the SNP and disease.

Calculate the **Expected** Affected risk alleles. Show your work

(Affected Total alleles x Total Risk alleles) / grand total alleles =



Calculate the **Expected** Unaffected risk alleles. Show your work

(Unaffected Total alleles x Total Risk alleles) / grand total alleles =

14. Interpret association significance. The app calculated a X^2 analysis to test the hypothesis that the risk allele frequency is the same in affected and unaffected individuals.

X^2 value: _____ p-value: _____

Answer the following question in the app:

Does this SNP appear associated with disease?

Select one answer

15. Run the genome-wide association scan. The app will run the same X^2 test for the remainder SNPs and provide a list of those with the highest disease association (i.e., lower p-values).

Finally, the Manhattan plot and associated table will indicate the SNPs and location (chromosomes) of those SNPs with the highest disease association

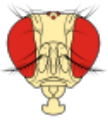
Which chromosome has the strongest association? _____

Which SNP has the strongest association? _____

Does this SNP necessarily cause the disease? Explain your answer.

***IMPORTANT NOTE*:**

Remember to return your assignment at the END of the lab session or to submit an electronic copy (i.e., scanned PDF file) through CANVAS



GENOME-WIDE ASSOCIATION STUDY (GWAS) WORKFLOW

